

TSE6223 SOFTWARE ENGINEERING FUNDAMENTALS

PROJECT GUIDELINES

Project Description

A major component of the course is a software development project. The objective is to develop a software application that could be used in a real-world scenario or to improve an existing system. Students may choose the type of application they wish to develop. The application can be a web application, mobile application or desktop application. The chosen type should be appropriate for the problem being addressed. Students can choose a programming language to develop the application, such as Python, PHP, or any other suitable language. The project team will work together through the full development cycle, from an initial feasibility study to delivering a functioning product. A key part of the project is to implement a software process model throughout the development and to document the software engineering processes involved. The group leader is responsible for coordinating the team, assigning tasks, and ensuring that all members contribute effectively to the project.

Title and Grouping

Choose a title from the list below. (You will not be able to change the title once the title is approved)

- Property Rental Management System
- MMU Student Enrollment System
- Employee Attendance System
- Asset Management System
- Job Application System

You must form a group of four members. All the members must be from the same lab section. The lecturer will randomly reorganise groups with fewer students.

Project Report

The report should be written in your own words (no copy-and-paste work). Your report should be reviewed by your lab instructor phase by phase. You may follow the timeline below.

No.	Topic	Initial Submission
1	Introduction and Background Study	Week 5
2	Requirements Engineering and Functional & Non-Functional Requirements	Week 5
3	Requirements Modelling	Week 7
4	Design Modelling	Week 9
5	Testing	Week 12
6	Project Management	Week 12

Submission Details

Your submission folder must include the following items:

- Project report (pdf) with cover page (see page 6), project declaration form, and similarity index report
- Source code
- Presentation slides (pdf)
- Presentation video (approximately 5 minutes for report and 5 minutes for system demonstration): Any content exceeding 10 minutes will not be assessed.

All members must participate in the presentation. The presentation plays a crucial role in justifying the implementation of your project and what you have accomplished. **Failure to participate in the presentation without a valid reason will be considered as having not completed the work, leading to a zero mark.**

You must submit the project before 26 June 2026 at 10.00 am. Each group should make only one submission. Your submission should be a single normal zipped file and name the zip file as LeaderName (e.g: AlanSmith.zip). It is your responsibility to ensure that your submission file is correct and can be opened without any errors. Please note that resubmission after the project deadline is not permitted for erroneous file submission.

MARKS WILL BE DEDUCTED FOR LATE SUBMISSION (1 MARK – 1 DAY).

PLAGIARISM: The similarity index of your report must not exceed 20%. Any report exceeding this limit will receive zero marks.

AI USAGE DECLARATION: Students must declare any use of AI tools and explain how they were used (e.g., for grammar checking or to help understand concepts) in the declaration form. failure to declare such use may be considered academic misconduct and could result in penalties. Your report will be checked for AI-generated content, and a high AI usage index may result in penalties.

See the next page for the marking scheme.

Marking Scheme

	0-1	2	3	4	5
A. Report (50 Marks)					
1. Introduction					
2. Background Study					
3. Requirements Engineering					
4. Functional & Non-Functional Requirements					
5. Requirements Modelling					
6. Design Modelling					
7. Testing					
8. Project Management					
9. Conclusion					
10. Quality of Report					
B. System (30 Marks)					
1. Functionality Completeness					
2. Functionality Accuracy					
3. User Interface (UI)					
4. User Experience (UX)					
5. Programming: implementation					
6. Programming: code quality					
C. Presentation (10 Marks)	M1	M2	M3	M4	
1. Understanding of subject matter					
2. Clarity, Professionalism and Confidence					
Total Marks 90					
Normalized Marks 30%					

See the next page for the details of the marking scheme.

Marking Scheme

Report (50 Marks)	Not/Minimally achieved	Needs improvement	Average	Good	Excellent
	0-1	2	3	4	5
Introduction: Provides an overview of the project, problem statements, objectives, and the significance of the system.					
Background Study: Provides essential context and foundational knowledge relevant to the subject matter and relevant system development.					
Requirements Engineering: Explains the steps undertaken in requirements engineering including inception, elicitation, elaboration, negotiation, specification, validation, and requirements management.					
Functional & Non-Functional Requirements: Provides functional and non-functional requirements of the system.					
Requirements Modelling: Provides a comprehensive analysis model of the system.					
Design Modelling: Provides a comprehensive design model of the system.					
Testing: Describes the testing techniques/strategies used, how the tests were conducted, and the test results.					
Project Management: Includes project planning, teamwork, risk and time management.					
Conclusion: Summarizes the key findings and insights.					
Quality of Report: Report is well-structured and formatted. List of references is provided.					

System (30 Marks)	Not/Minimally achieved	Needs improvement	Average	Good	Excellent
	0-1	2	3	4	5
Functionality Completeness: All required functions and features are implemented, meeting the system's requirements.					
Functionality Accuracy: The functions and features perform as expected and produce the desired outcomes.					
User Interface (UI): The visual design and interactive elements of the system, including layout, buttons, icons, and typography.					
User Experience (UX): The usability, efficiency and accessibility of the system.					
Programming: implementation: The coding accurately implements the required functions and features.					
Programming: code quality: The readability, efficiency, maintainability, and adherence to coding standards or best practices in the code.					

Presentation (10 Marks)	Member 1	Member 2	Member 3	Member 4
Understanding of subject matter: Demonstrates strong understanding of the subject matter with clear explanations. (0-1: Not/Minimally achieved; 2: Needs improvement; 3: Average; 4: Good; 5: Excellent)				
Clarity, Professionalism and Confidence: Speech is clear and fluent. Professional in tone, attire, and body language. Demonstrates strong confidence. (0-1: Not/Minimally achieved; 2: Needs improvement; 3: Average; 4: Good; 5: Excellent)				



Faculty of Information Science & Technology

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Trimester March/April 2026

Project Report

Title:

Student ID	Student Name	Programme